

Research on failure prediction of wind turbines in offshore wind clusters

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Abstract

Offshore wind is a very promising renewable energy source, and offshore wind power has been widely used. However, the maintenance of offshore wind power is a very inconvenient task, which not only is difficult to operate but also can be very costly, and it is essential to anticipate offshore wind power failures in advance. In this paper, we focus on wind turbine faults for analysis, wherein we use a high-performance radial basis function (RBF) neural network, and to obtain the center of its RBF, the normalization constant and the weighting coefficients from the hidden layer to the output layer for better performance, we use an optimized particle swarm optimization algorithm so as to obtain more accurate parameters and therefore seek better performance. We focus on two very high failure rates of wind turbines for testing, and the results show that our proposed method can predict the occurrence of failures 10–20 h in advance and thus have a good performance.

Keywords: offshore wind; RBF neural network; PSO; Wind turbine failure prediction

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1 INTRODUCTION

As a renewable energy source, wind energy is non-polluting and sustainable to the environment, and it is currently receiving strong attention worldwide [1], but the correlation between wind energy resources and the environment is high; for example, in coastal areas, mountainous areas and plain land areas, wind energy resources are different, with offshore wind energy resources being 20% to 40% higher compared to land [2, 3].

In the early 1990s, Europe has begun to study offshore wind power; after almost 30 years of development, the global installed capacity of offshore wind power is growing, but the resulting operation and maintenance problems are gradually increasing [4], such as high failure rate (the working environment is harsher compared to onshore wind farms) [5], poor accessibility (after a failure often due to weather factors, wherein maintenance personnel cannot reach the location of the failure on time) [6] and the difficulties of operation and maintenance (large engineering vessels are required to assist personnel who need to go to the sea for maintenance) [7]. The ability to predict offshore wind power plant failures in advance and then develop maintenance and repair

plans is of great importance to the economics of offshore wind power plant operation [8].

In recent years, offshore wind power has gradually developed in the direction of more refinement in the improvement of wind turbine equipment and troubleshooting arrangements [9]. Arshad and O'Kelly [10] took wind turbine fault diagnosis as the object and summarized the existing diagnostic methods for wind turbines. With the rapid development of artificial intelligence and deep learning, data-driven fault diagnosis methods have become a research hotspot in the field of wind power nowadays [11]. Data-driven fault diagnosis methods are the mainstream methods used for fault diagnosis because they do not require proficiency in system principles, are easy to model and can be automated [12]. These include artificial neural network, rough set, support vector machines (SVM), fuzzy mathematics, artificial immune systems, back propagation (BP) neural network, radial basis function (RBF) neural network, etc.; the excellent performance of the RBF neural network has been verified in many fields [13], and it is very effective to predict the failure of offshore wind turbines. However, the three parameters of the neural network (the center of the basis function, the variance and the weights from the implicit

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layer to the output layer) greatly affect its performance [14]. To better perform the RBF neural network, the optimal parameters need to be sought, which requires a better optimization-seeking algorithm. The particle swarm optimization (PSO) algorithm as a classic optimization-seeking algorithm has been widely used [15], and its performance has been widely proven, and it is very reliable to optimize RBF with it.

Previous studies [16–19] relied on genetic algorithms to optimize the calculation of BP neural networks and accurately diagnosed many conditions of normal or abnormal generator operation. Zhang *et al.* [20] studied various types of faults in gearboxes and applied the PSO to optimize the SVM algorithm to accurately identify and diagnose various types of faults in gearboxes in wind turbines. Yang *et al.* [21] studied a data analysis-based fault diagnosis method for wind turbines, in which the quantities that may be related to wind power faults were selected from the supervisory control and data acquisition (SCADA) system monitoring quantities and transformed into a fault sample transaction set, and then the pattern of fault development was judged.

In this paper, we will analyze two very typical faults in wind turbines and use the optimized PSO to obtain the key parameters of the RBF neural network to obtain a better learning effect and then predict the faults in advance.

2 PROPOSED STRATEGY

2.1 RBF neural network

RBF neural network is a local approximation feedforward neural network that can approximate arbitrary nonlinear functions and has good generalization ability when dealing with regularities that are difficult to resolve within the system. Since the middle implicit layer of RBF neural network has only one layer, it has fast learning and convergence speed compared with the BP neural network, and the implicit layer uses the Euclidean distance between the input vector and the center vector as the self-variable, and the Gaussian function as the activation function; when the input is farther from the center of the activation function, the output value of the implicit layer is smaller, so only when the Euclidean distance between the input and the center is smaller. Therefore, the real mapping effect only occurs when the Euclidean distance between the input and the center is smaller, so it has the property of local approximation. Liu *et al.* [22] has demonstrated that generalized RBF neural networks have superior continuous function approximation performance and high noise immunity.

Figure 1 shows the topology of a single-output RBF neural network. The input and output of the hidden layer of the RBF neural network are shown in Equations (1) and (2).

$$h_j = \exp\left(-\frac{\|x - c_j\|^2}{2\sigma_j^2}\right) \quad (1)$$

$$y = \sum_{j=1}^s \omega_j \times h_j \quad (2)$$

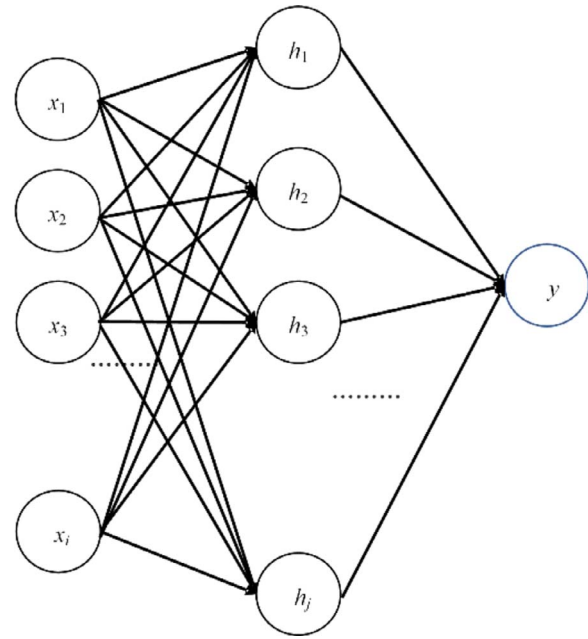


Figure 1. Single-output RBF neural network topology.

where c_j is the center vector of the j th node of the hidden layer, $\|x - c_j\|$ is the Euclidean distance between the input vector and the center vector, σ_j is the width of the j th node in the hidden layer, h_j is the output of the j th node in the hidden layer, ω_j is the weight of the j th neuron and y is the output of the network.

There are three main parameters for solving in RBF networks: the center of the basis function, the variance and the weights from the implicit layer to the output layer. First, the center and variance of the implicit layer basis functions are solved through an unsupervised learning process. Then, the weights between the hidden layer and the output layer are solved by a supervised learning process.

The b k -means clustering centers are selected, and the variance σ_j is solved by Equation (3).

$$\sigma_j = \frac{c_{\max}}{\sqrt{2b}}, j = 1, 2, \dots, b \quad (3)$$

where $c_{\max}$ is the maximum distance between the selected center points. The connection weights of the neurons between the implicit layer and the output layer can be directly calculated by the least squares method, which can be reduced to obtain Equation (4).

$$\omega_j = \exp\left(\frac{b}{c_{\max}^2} \|x - c_j\|^2\right) \quad (4)$$

Although RBF neural networks can handle nonlinear problems well, the determination of the parameters of the neural network is closely related to the output of the model. Therefore, the determination of several parameters, such as the center of RBF,

normalization constant and the weighting factor from the hidden layer to the output layer, is particularly important.

2.2 PSO optimization RBF

The main idea of the PSO is to simulate the behavior of a flock of birds foraging, where the position of each particle represents a solution, and after each iteration, each particle converges to the individual historical optimal solution. Assuming that the objective is solved in M -dimensional space $X_i = (X_{i1}, X_{i2}, \dots, X_{iM})$, $V_i = (V_{i1}, V_{i2}, \dots, V_{iM})$ and $P_i = (P_{i1}, P_{i2}, \dots, P_{iM})$ represent the spatial position, flight speed and best position of the i th particle in the whole population, respectively. The global optimal position of the whole particle population is $G = (G_1, G_2, \dots, G_M)$.

The algorithm's speed and position update are shown in Equations (5) and (6).

$$V_{jd}(t+1) = wV_{jd}(t) + c_1r_1(P_{jd}(t) - X_{it}(t)) + c_2r_2(P_{gt}(t) - X_{it}(t)) \quad (5)$$

$$X_{id}(t+1) = X_{id}(t) + V_{id}(t+1) \quad (6)$$

where d is the dimensionality of the population, i is the population size, t is the number of iterations, w is the weight factor, c_1 and c_2 are learning factors, r_1 and r_2 are random numbers located in the middle of (0,1) and V_{id} is the maximum velocity.

As shown in Equations (5) and (6), the size of the three parameters w , c_1 and c_2 affects the particle velocity and position, which directly affects the global search capability and convergence speed of the algorithm. It was demonstrated in [23] that a larger w at the early stage of the search can be searched in a larger search range, and a smaller w at the later stage can be achieved in a local precise search. In this paper, we design a segmented nonlinear decreasing approach. If the value of self-learning factor c_1 is too large, it will make the particles wander in the local range and the global search ability is insufficient. If the value of social learning factor c_2 is too small, the particles will converge to the local area too early, as shown in Equation (7),

$$w(t) = \begin{cases} w_{\max} & t \leq \text{CNT}/2 \\ \frac{A}{t-B} (w_{\max} - w_{\min}) & t > \text{CNT}/2 \end{cases} \quad (7)$$

where

$$A = \frac{w_{\max} \times w_{\min} \times \text{CNT}}{2(w_{\max} - w_{\min})^2} \quad (8)$$

$$B = \frac{\text{CNT}}{2} \left(1 - \frac{w_{\min}}{w_{\max} - w_{\min}} \right) \quad (9)$$

where $w_{\max}$ is the upper limit of inertia weight, $w_{\min}$ is the lower limit of inertia weight, CNT is the maximum number of iterations and t is the current number of iterations.

Combining the given characteristics, as shown in Equations (10) and (11), c_1 is linearly decreasing and c_2 is linearly increasing.

$$c_1 = c_{1\max} - \frac{t(c_{1\max} - c_{1\min})}{\text{CNT}} \quad (10)$$

$$c_2 = c_{2\min} + \frac{t(c_{1\max} - c_{1\min})}{\text{CNT}} \quad (11)$$

where $c_{1\max}$ is the maximum value of the i th learning factor and $c_{1\min}$ is the minimum value of the i th learning factor.

The method ensures that the global search is completed at a high speed when the particle is initially searched, and the nonlinear decrement is performed in the middle, which ensures the fine search at the later stage and improves the disadvantage that the direct linear decrement cannot guarantee the completion of the global search; at the same time, the nonlinear characteristics of the optimization search are used in the particle nonlinear space, which is conducive to jumping out of the local optimum. As shown in Equations (10) and (11), in the process of searching for the best, c_1 takes a larger value and c_2 takes a smaller value at the early stage of the search, which makes the particles ensure the search speed at the early stage to fully search globally; at the later stage of the search, c_1 becomes smaller and c_2 becomes larger to enhance the local search ability.

The principle of the optimized RBF neural network is to map the parameters as particle objectives in PSO and return to the RBF neural network after finding the optimal solution using the PSO search function to realize the construction of the network model. In the process of searching, the one with the smallest mean square error is used as the fitness function, and the weight is optimal when it is the smallest.

Figure 2 shows the flow chart of the proposed strategy in this paper. PSO is applied to seek the three key parameters affecting the RBF neural network, which, as analyzed earlier, greatly affect the prediction performance of the RBF, and the optimized RBF will be applied to the fault prediction of offshore wind power generation to obtain better performance.

3 MAIN FAULT TYPES IN WIND TURBINE FAILURE

Because offshore wind turbines have been in a harsher working environment, the installation height must also be more than 80 m. Therefore, as the working time grows, the frequency of accidents increases accordingly, and there are various types of failures. The main components inside the wind turbine, such as the gearbox, generator and main bearings, are highly susceptible to damage when working in adverse environments.

Stetco et al. [24] statistically compared the failure rates of major wind turbine devices in 5 to 10 offshore wind farms and found that the frequency of failure of components such as pitch systems, gearboxes and generators was high, as shown in Figure 3.

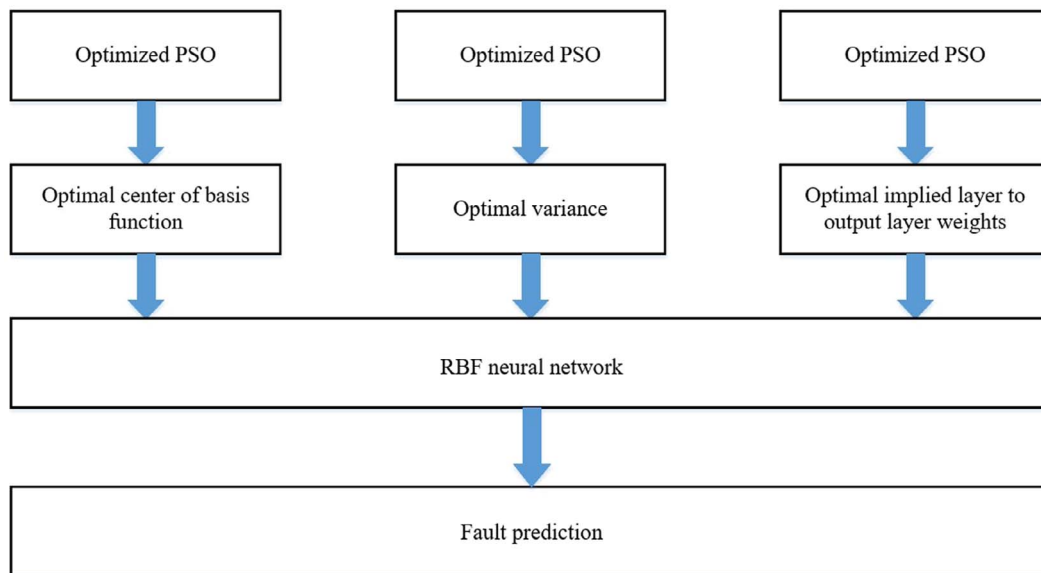


Figure 2. The flow chart of the proposed strategy.

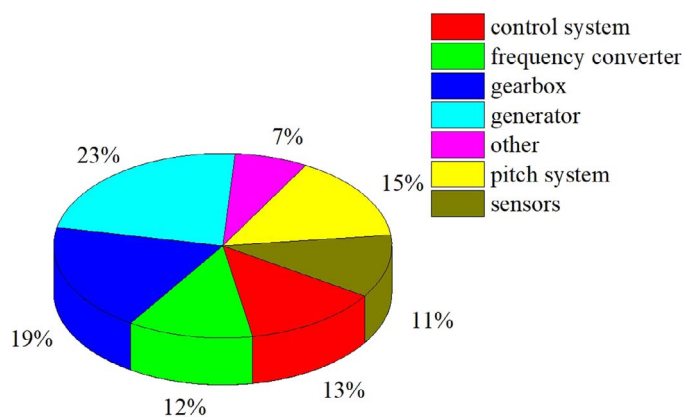


Figure 3. Main fault statistics.

As shown in Figure 3, the component with the highest frequency of wind farm failures is the generator, followed by the gearbox, so they will be selected as the object of study in this paper.

3.1 Generator

Generator failure statistics are shown in Figure 4.

As shown in Figure 4, the highest frequency of faults is the bearing over temperature fault, about 54%, which is much higher than other generator faults, so it is selected for study in this paper.

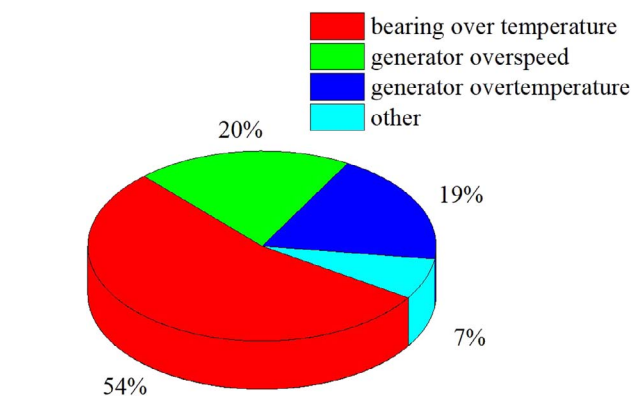


Figure 4. Generator failure statistics.

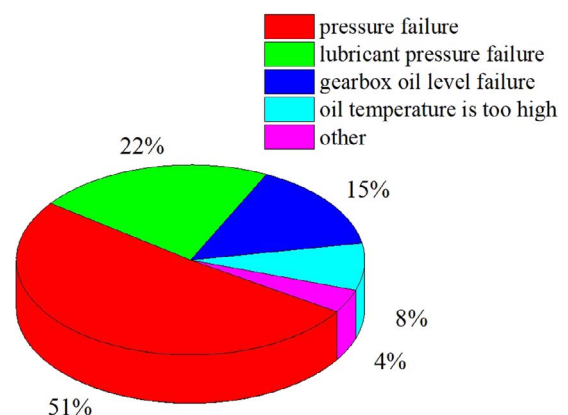


Figure 5. Gearbox failure statistics.

3.2 Gearbox

Gearbox failure statistics are shown in Figure 5.

As shown in Figure 5, the highest frequency of gearbox failure is pressure failure, about 51%, and considering that high frequency failure is the most representative, it was selected for this paper.

There are so many faults in offshore wind farms that we cannot list, and we cannot test all of them in this paper, but as shown in Figures 3–5, generator and gearbox faults are very typical in

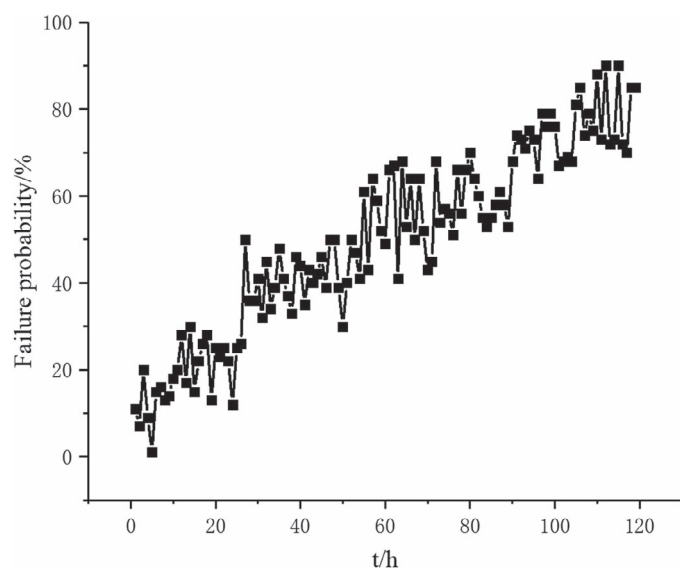


Figure 6. Results of case 1.

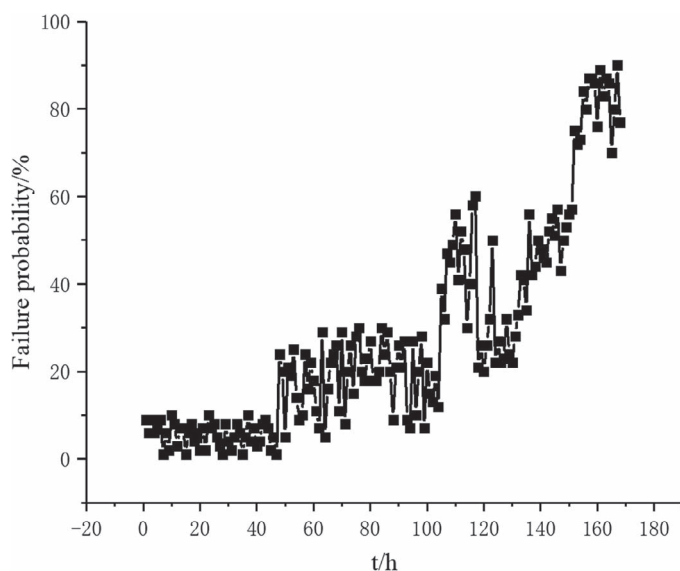


Figure 7. Results of case 2.

offshore wind farms because they account for a very high percentage of faults, and among them, the bearing over temperature fault and pressure failure can be considered as typical faults, so this paper uses these two typical faults for detailed analysis instead of using all faults, because it is impractical to use all faults.

4 RESULTS AND DISCUSSION

We used the data we collected to optimize the RBF neural network parameters using the optimized PSO and then trained the RBF neural network algorithm. The goal of the algorithm was to seek faults in the wind turbine based on the current data. For the wind

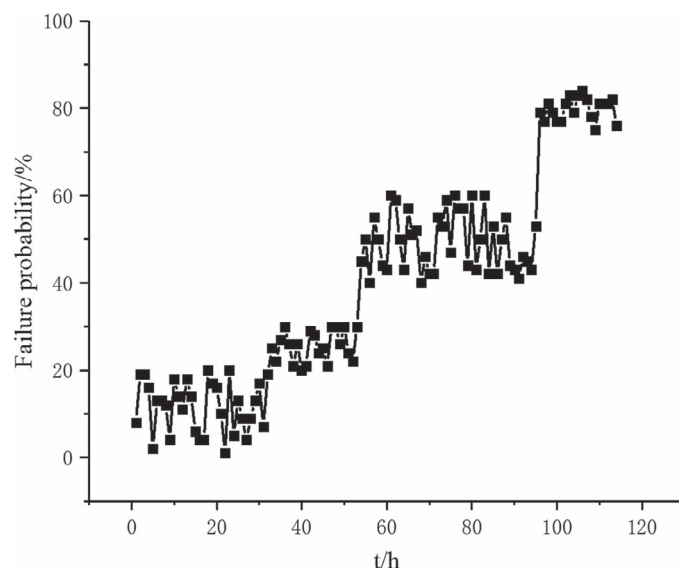


Figure 8. Results of case 3.

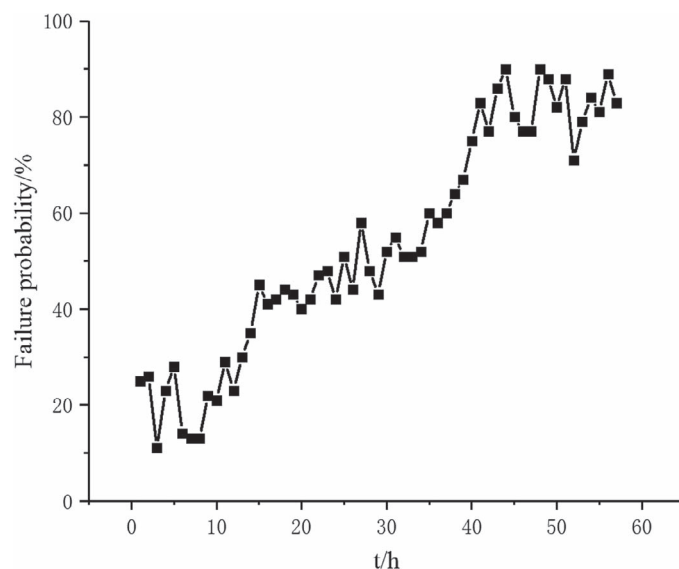


Figure 9. Results of case 4.

turbine faults, we focused on only two points, the bearing over temperature fault (generator) and the pressure failure (gearbox), which as we described in Section 3 are very frequent and very representative, and we used four sets of faults to test the effectiveness of the algorithm.

4.1 Case 1

No. 5 of the wind farm reported the bearing over temperature fault in March 2016, and the SCADA data were imported into this paper's model for prediction, and the resulting failure probability are shown in Figure 6.

In the SCADA data fault log, the first fault alarm is issued by the system at about 120 h. As shown in Figure 6, after about 60 h, the RBF neural network determines that the probability of the bearing over temperature fault is more than 50%, and after 100 h, the probability of fault is nearly 80%, so it is almost assumed that there is an overtemperature fault in the wind turbine; the model in this paper predicts the fault about 20 h in advance, and the effect is good.

4.2 Case 2

Figure 7 is wind farm No. 13, which reported in May 2018 the bearing over temperature fault; the same as case 1, we import SCADA data for analysis. In the SCADA data fault log, the first fault alarm is issued by the system at about 170 h. As shown in Figure 7, after about 160 h, the RBF neural network determines that the probability of the bearing over temperature fault is more than 80%, so it is almost assumed that there is an overtemperature fault in the wind turbine; the model in this paper predicts the fault about 10 h in advance.

4.3 Case 3

Regarding gearbox pressure failure, wind farm No. 7 reported this failure in October 2014, and the probability after importing SCADA data is shown in Figure 8.

In the SCADA data fault log, the first fault alarm is issued by the system at about 120 h. As shown in Figure 8, after about 100 h, the RBF neural network determines that the probability of the bearing over temperature fault is more than 80%, so it is almost assumed that the wind turbine will suffer from the gearbox pressure failure; the model in this paper predicts the fault about 20 h in advance.

4.4 Case 4

No. 16 reported a gearbox pressure failure in January 2017, and the probabilities after importing the SCADA data are shown in Figure 8. In the SCADA data fault log, the first fault alarm is issued by the system at about 60 h. As shown in Figure 9, after about 40 h, the RBF neural network determines that the probability of the gearbox pressure failure is more than 80%, so it is almost assumed that the wind turbine will suffer from gearbox pressure failure; the model in this paper predicts the fault about 20 h in advance.

As shown in Figures 6–9, when the failure probability rises to about 80%, we consider that the failure has occurred because, as a rule of thumb, 80% or more is a more appropriate value. The reason why it is not 100% is that our training data are not large enough and there may be many scenarios that have not been fully considered, and it is extremely unreasonable to use 100% for industrial applications; when we have enough training data, perhaps the probability of failure will rise above 90%, and we will decide that a failure has occurred.

5 CONCLUSION

In this paper, we use RBF neural networks and an optimized PSO algorithm for analysis, which is optimized to obtain more accurate parameters needed for RBF neural networks. The experimental results of the imported SCADA test data with the proposed method in this paper show that the optimized RBF neural network has satisfactory performance, and it has good application prospects to predict the failure of offshore wind power generation in advance.

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AUTHOR CONTRIBUTIONS

Wenjin Chen: Data curation, Methodology, Writing – original draft, Writing – review & editing; Hao Huang: Formal analysis, Investigation, Supervision; Jun Zhang: Funding acquisition; Silei Yao: Formal analysis, Supervision, Writing – original draft; Ruoyi Zhang: Writing – review & editing.

DATA AVAILABILITY

The data that support the findings of this study are available upon reasonable request from the authors.

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